Using Normalizing Flows for Weighting Estimators with Continuous Treatment

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Motivation

- Causal inference under **unconfoundedness** relies on accurate propensity score estimation
- AIPW estimator is **doubly robust** and semiparametrically efficient
- **Continuous treatment** is common in political science:
  - Amount of foreign aid on economic development
  - Campaign spending on vote share
  - Duration of peacekeeping missions on conflict recurrence
  - Level of media exposure on political attitudes
- Two challenges when treatment is continuous:
  - Dichotomizing loses information on treatment effect heterogeneity
  - No logit/probit analogue — need to estimate a *conditional density* $f(T \mid X)$
- Existing approach: kernel density estimation (KDE)
  - Sensitive to kernel choice and bandwidth
- **Our proposal:** use *Normalizing Flows* (NF) for propensity score estimation with continuous treatment

AIPW with Continuous Treatment

- Under unconfoundedness $T \perp\!\!\!\perp Y(t) \mid X$ and overlap, the **dose–response function** is:

$$m(t) = \mathbb{E}[Y(t)] = \mathbb{E}\left[\mu(t, X) + \frac{K_h(T - t)}{f(T \mid X)}\left(Y - \mu(T, X)\right)\right]$$

  - $\mu(t, x) = \mathbb{E}[Y \mid T = t, X = x]$: outcome model
  - $f(T \mid X)$: **generalized propensity score** (GPS)
  - $K_h(\cdot)$: kernel with bandwidth h

Double Robustness

Consistent if *either* the outcome model μ or the GPS $f(T \mid X)$ is correctly specified (Robins and Ritov, 1997).

Normalizing Flows

- **Idea:** transform a simple base distribution (e.g., Gaussian) into a complex target via invertible maps.
- **Change of variables:**

$$p_{\mathbf{Y}}(\mathbf{y}) = p_{\mathbf{Z}}(\mathbf{f}(\mathbf{y}))\,|\det \mathrm{D}\,\mathbf{f}(\mathbf{y})|$$

  - **f**: invertible transformation
  - Exact density evaluation & sampling
  - No kernel/bandwidth choices needed

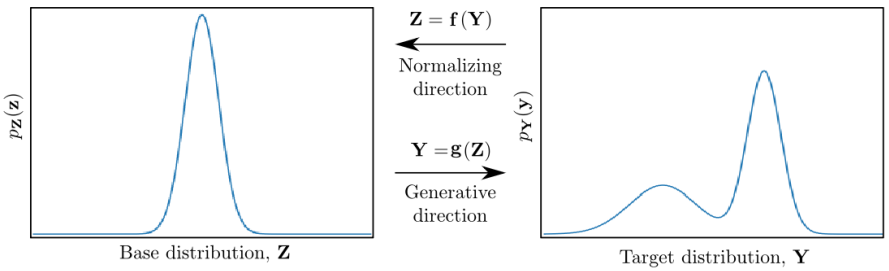

Figure: From Kobyzev et al. (2021)

NF for Propensity Score Estimation

- **Goal:** estimate the conditional density $f(T \mid X)$ using conditional NF
  - Train a conditional NF: transformation parameters depend on covariates X
  - Yields **exact** conditional density $\hat{f}(T \mid X)$ via change of variables
  - Plug $\hat{f}(T \mid X)$ into the AIPW estimator as the GPS
- **Advantages over KDE:**
  - No kernel/bandwidth choices
  - Captures complex, multimodal densities
  - Scalable to higher-dimensional covariates

Figure: Conditional two-moons: true density (left) vs. NF estimate (right)

Simulation Results

- Compare DR estimators using: (1) true GPS, (2) residual KDE, (3) NF
  - NF-based DR estimator performs **comparably to true GPS**
  - Bias and RMSE decrease with n

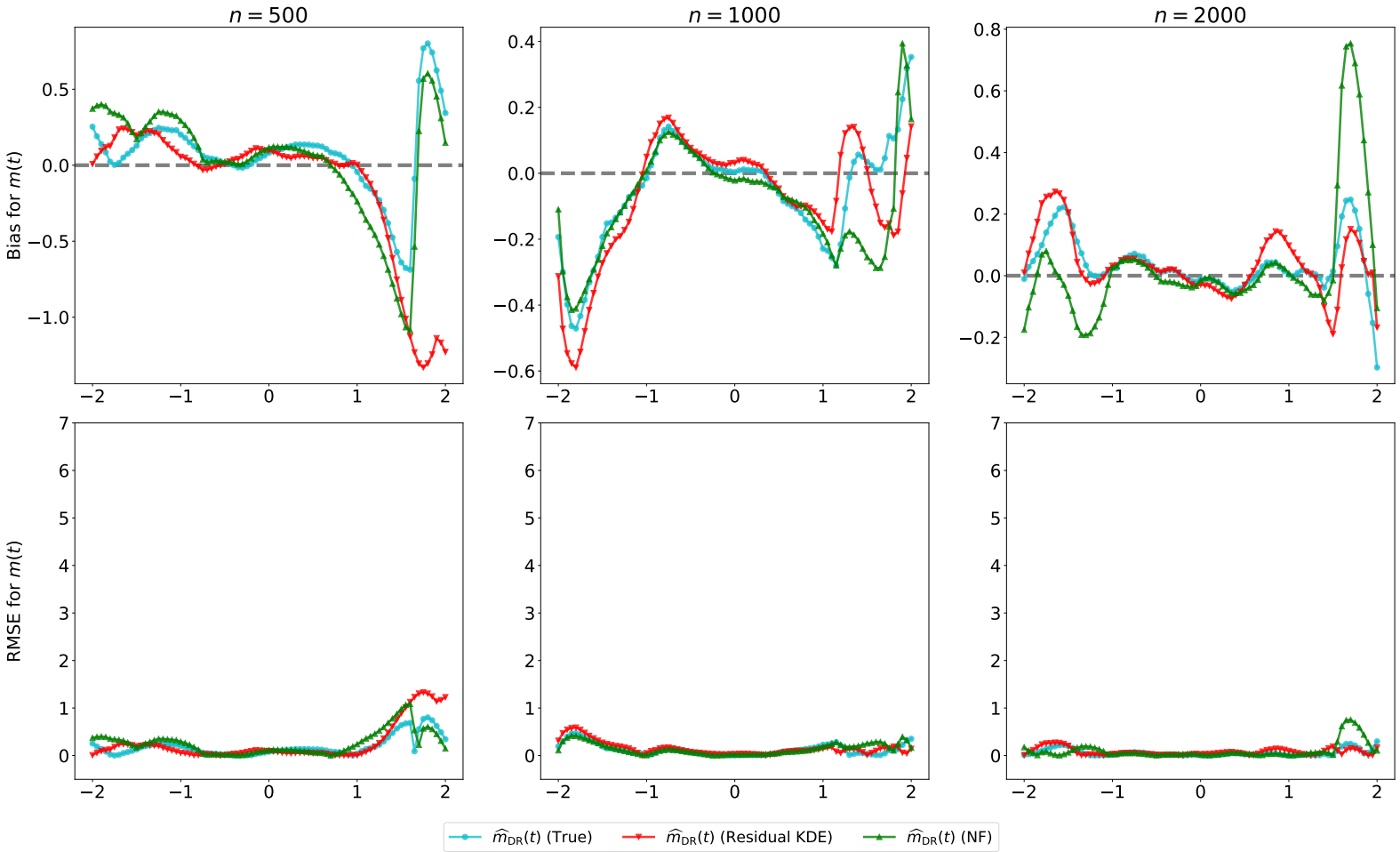

Figure: Bias and RMSE for dose–response $m(t)$ across sample sizes ($n = 500, 1000, 2000$)

Conclusion & Next Steps

Summary:

- NF provides accurate, tuning-free GPS estimation for continuous treatments
- NF-based AIPW estimator achieves comparable performance to oracle (true GPS)
- Avoids kernel/bandwidth sensitivity of KDE

Next steps:

- Theoretical properties of NF-based GPS estimator
- Application to political science data with continuous treatment
- Comparison with other ML-based GPS estimators (e.g., random forest)

References I

KOBYZEV, I., S. J. PRINCE, AND M. A. BRUBAKER (2021): "Normalizing Flows: An Introduction and Review of Current Methods," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43, 3964–3979.

ROBINS, J. M. AND Y. RITOV (1997): "Toward a Curse of Dimensionality Appropriate (Coda) Asymptotic Theory for Semi-Parametric Models," *Statistics in Medicine*, 16, 285–319.